

CARE CONNECT

Smart Healthcare Management & AI-Assisted Clinical Appointment Platform

Academic Project:	Final Year College Capstone Project
Degree Program:	Bachelor of Technology / B.E. in Computer Science & Engineering
Academic Year:	2025 – 2026
Primary Domain:	Healthcare Informatics, Web Systems, Applied AI, Data Security
Core Tech Stack:	Python 3.14, Flask, SQLite, SQLAlchemy ORM, ReportLab, Scikit-Learn
Test Verification:	263 / 263 Tests Passed (100% Pass Rate, 0 Failures, 0 Errors)
Database Integrity:	SQLite Production Database Intact (database/careconnect.db preserved)
Documentation Date:	September 2026

Project Scope & Statement:
Care Connect is a complete, role-based healthcare web application built to streamline doctor-patient appointments, eliminate double-booking race conditions through database-level partial unique constraints, deliver tamper-resistant ReportLab PDF passes and Zero-PHI QR code digital verification, and provide non-diagnostic AI operational assistance.

2. Abstract

In modern healthcare ecosystems, patients frequently encounter difficulties identifying appropriate medical specialists, navigating fragmented booking systems, and enduring prolonged waiting periods. Simultaneously, healthcare facilities face scheduling bottlenecks, double-booking collisions, and challenges in operational workload forecasting. **Care Connect** is a centralized, role-based healthcare web application developed to bridge the communication and workflow gap between patients, medical practitioners, and hospital administrators.

Built using Python 3.14, Flask, and SQLAlchemy over an SQLite relational database, the system enforces strict Role-Based Access Control (RBAC) across three primary user personas: Hospital Administrators, Doctors, and Patients. The application incorporates transactional appointment booking with 30-minute dynamic slot generation and database-level double-booking protection. Following appointment confirmation, patients receive a tamper-resistant PDF appointment pass generated via ReportLab and a secure digital QR code verification gateway that exposes Zero Protected Health Information (Zero-PHI) to public scanners.

Care Connect integrates four explainable, ethically aligned AI modules designed strictly for administrative and navigational guidance: AI Specialist Recommendation, AI Patient Load Prediction, AI Smart Appointment Recommendation, and the Care Connect AI Assistant. The platform is fortified with defense-in-depth security, including Scrypt password hashing, CSRF protection, IDOR defenses, HTTP security headers, and an accessible, responsive dual Light and Dark theme interface. All 263 automated test cases pass with 100% success rate with zero warnings, confirming system reliability.

3. Introduction

Outpatient management is the primary entry point for non-emergency clinical care. Traditional scheduling mechanisms—such as telephone inquiries, physical reception queues, and disjointed hospital registries—introduce high administrative overhead, human error, and scheduling conflicts. Care Connect provides a unified, web-accessible healthcare portal designed to streamline interactions between patients, doctors, and healthcare institutions. The project emphasizes transparent scheduling, mathematical appointment load forecasting, and ethical AI deployment where artificial intelligence assists operational workflows without making diagnostic or prescriptive clinical decisions.

4. Problem Statement

Outpatient healthcare workflows are hindered by five critical challenges:

- **Fragmented Provider Information:** Patients lack unified visibility into registered hospitals, practicing specialists, consultation fees, and real-time availability.
- **Specialist Misdirection:** Patients without clinical knowledge frequently book appointments with inappropriate departments, delaying effective care.
- **Double-Booking Race Conditions:** Concurrent user requests frequently lead to double-booked slots when software lacks database-level concurrency protection.
- **Unverified Receipts & PHI Leaks:** Paper slips are easily lost, while digital passes often leak private phone numbers, home addresses, and clinical symptoms.
- **Unpredictable Patient Flow:** Hospital administrators struggle to forecast outpatient volumes, leading to inefficient staffing and resource bottlenecks.

5. Objectives

1. Engineer a modular, multi-role Flask web application providing isolated environments for Hospitals, Doctors, and Patients.
2. Eliminate double-booking collisions through dynamic 30-minute slot generation and database engine-level partial unique constraints.
3. Provide automated ReportLab PDF appointment passes and cryptographically secure QR code digital verification.
4. Enforce strict Zero-PHI privacy standards on the public QR verification gateway.

5. Deploy explainable AI assistance for specialist selection, patient load forecasting, smart slot scoring, and platform Q&A.;
6. Ensure enterprise-grade security including Script password hashing, CSRF tokens, XSS auto-escaping, and HTTP security headers.
7. Deliver an accessible, dual-theme (Light/Dark) design system compliant with WCAG AA guidelines.

6. Existing System

Existing outpatient workflows rely on manual reception desks, telephone switchboards, and basic calendar applications. These systems suffer from human logging errors, lack of real-time schedule synchronization, absence of specialist guidance, inability to forecast patient volumes, and severe security vulnerabilities due to unencrypted storage of medical notes.

7. Proposed System

Care Connect introduces an integrated, transaction-safe digital healthcare platform. The system features dedicated role dashboards, real-time physician schedule synchronization, database-enforced booking integrity, offline-verifiable PDF/QR passes, four non-diagnostic AI operational engines, and complete multi-tenant hospital data isolation.

8. System Architecture

Care Connect follows the Model-View-Controller (MVC) architectural pattern implemented through Flask's modular Blueprint framework. The architecture is organized into four distinct tiers: Client Layer, Security & Application Layer, Business & AI Services, and Data Persistence.



9. Technologies Used

Component	Technology	Version	Role in Care Connect
Backend Framework	Flask	3.1.3	Application factory, blueprint modular routing, middleware
Database Engine	SQLite	3.x	Local relational data storage; preserved database/careconnect.db
ORM Layer	Flask-SQLAlchemy	3.1.1	Object-relational mapping, transactional sessions, migrations
Authentication	Werkzeug Security	3.1.8	Cryptographic Scrypt / PBKDF2 password hashing
Document Engine	ReportLab	5.0.1	Programmatic appointment confirmation PDF generation
QR Code Engine	qrcode + Pillow	8.2 / 12.3	High-res 2D barcode generation encoding UUID4 tokens
Data Science / ML	Scikit-Learn / Pandas	1.9.1 / 3.0.5	Time-series data aggregation, moving average forecasting
UI Styling & Theme	Bootstrap 5.3 + CSS3	5.3.0	Responsive grid, custom variables for Light/Dark mode

10. User Roles

Care Connect implements three distinct, isolated authenticated roles:

- **Hospital Administrator (role='hospital')**: Manages institutional metadata, doctor onboarding, staff rosters, facility bed capacity, operational notices, and patient-load forecasts. Strictly prohibited from accessing patient booking or doctor availability pages.
- **Doctor / Specialist (role='doctor')**: Configures practicing working days and consultation hours, views daily appointment queues, reviews patient history, marks consultations completed, and queries the AI Assistant. Restricted from hospital admin or booking appointments as a patient.
- **Patient (role='patient')**: Discovers registered hospitals, filters doctors by specialty, accesses AI recommendation tools, books appointments, downloads PDF passes, verifies QR tokens, and manages booking history. Restricted from hospital and doctor portals.

11. Hospital Management Module

Accessible at `/hospital/dashboard`, this module provides administrative governance over a healthcare facility. Features include facility profile management (registration number, bed count, daily alert banner), a doctor staff roster allowing addition and editing of doctors and consultation fees, a master appointments queue displaying all visits scheduled at the hospital, and multi-tenant isolation ensuring Hospital A cannot access Hospital B's data.

12. Doctor Management Module

Accessible at `/doctor/dashboard`, this module supports clinical schedule management. Doctors configure their weekly practicing days and daily start/end consultation hours via `/doctor/availability`. The appointment queue categorizes patient visits into **Today's**, **Upcoming**, and **Completed** tabs. Doctors can mark active appointments as `'completed'`, which finalizes the consultation and locks the record against subsequent cancellation.

13. Patient Module

Accessible at `/patient/dashboard`, this module provides an intuitive health portal for patients. Patients can view upcoming appointment alerts, search registered healthcare facilities, obtain AI-assisted specialty and slot guidance, track appointment status, and cancel scheduled visits with automatic slot release.

14. Hospital Search and Doctor Discovery

The public and patient directory at `/patient/hospitals` lists all registered healthcare facilities with real-time bed capacity and notices. Clicking a facility opens `/patient/hospitals/`, displaying affiliated physicians categorized by medical specialty (Cardiology, Neurology, Pediatrics, Dermatology, Orthopedics, General Medicine) with consultation fees and direct booking links.

15. Appointment Booking System

Care Connect implements an 8-step visual booking pipeline: 1. *Hospital* $\rightarrow$ 2. *Specialization* $\rightarrow$ 3. *Doctor* $\rightarrow$ 4. *Date* $\rightarrow$ 5. *Slot* $\rightarrow$ 6. *Details* $\rightarrow$ 7. *Review* $\rightarrow$ 8. *Confirm*. The booking engine dynamically calculates 30-minute consultation intervals (`'09:00 AM - 09:30 AM'`) based on the physician's hours. Double-booking is prevented at the database engine level via SQLite's partial unique index (`'uq_doctor_date_slot'`), ensuring that concurrent transactions attempting to book the same slot are rejected with an `IntegrityError`.

16. Appointment Cancellation and History

Patients can review their complete consultation history at `/patient/appointments`, organized into *Upcoming*, *Today's*, *Completed*, and *Cancelled*. Active upcoming appointments can be cancelled via a confirmation modal. Cancellation updates the status to `cancelled`, immediately releasing the slot so another patient can book it. Completed and cancelled appointments cannot be cancelled again.

17. PDF Appointment Confirmation

Following booking, patients can download a tamper-resistant digital appointment pass generated dynamically via ReportLab (`utils/pdf_generator.py`). The document includes official Care Connect Deep Teal branding, the appointment tracking number (`CC-YYYY-XXXXXX`), patient demographics, hospital address, doctor credentials, scheduled time window, consultation fee, and an embedded high-resolution QR code. The file is streamed directly in-memory as `application/pdf`. Strict IDOR checks prevent patients from downloading passes belonging to others.

18. QR Verification (Zero-PHI Public Gateway)

Each appointment generates a cryptographically secure 128-bit URL-safe verification token stored in `Appointment.verification_token`. Scanning the QR code opens the public gateway at `/verify/appointment/` without requiring user authentication. To protect patient privacy, the gateway displays **Zero Protected Health Information (Zero-PHI)**:

Information Category	Public QR Gateway Status	Rationale & Privacy Justification
Appointment Status	VERIFIED / DISPLAYED	Confirms validity of appointment for security/check-in
Tracking Number	VERIFIED / DISPLAYED	Unique reference formatted as CC-YYYY-XXXXXX
Hospital Facility	VERIFIED / DISPLAYED	Identifies correct medical center location
Doctor & Specialty	VERIFIED / DISPLAYED	Directs patient to the assigned consultation room
Date & Time Slot	VERIFIED / DISPLAYED	Validates scheduled consultation arrival window
Patient Name / Phone	STRICTLY WITHHELD	Protects identity if physical pass is lost or misplaced
Symptoms / Notes	STRICTLY WITHHELD	Confidential clinical notes accessible only to physician

19. AI Specialist Recommendation

Implemented in `ai/specialist_recommender.py`, this module evaluates patient-described concerns against an indexed clinical vocabulary spanning Cardiology, Neurology, Dermatology, Orthopedics, Pediatrics, Ophthalmology, ENT, and General Medicine. The engine calculates term-matching density, provides explainable rationale tags, displays confidence levels, and displays verified physicians from the database who practice that specialty.

MANDATORY AI CLINICAL BOUNDARY NOTICE:

All Care Connect AI modules function strictly for administrative, logistical, and scheduling assistance. They are **NOT** medical diagnosis systems, do not prescribe medications, and do not provide medical treatment instructions. Every AI interface prominently displays: *'This recommendation is for appointment guidance only and is not a medical diagnosis.'*

20. AI Patient Load Prediction

Implemented in `ai/patient_load_predictor.py`, this module forecasts expected 7-day appointment volumes for hospital administration. It queries historical visits strictly for the requesting facility, excludes cancelled slots, and applies a 7-day weighted moving average combined with day-of-week seasonality multipliers. This operational tool helps hospitals allocate nursing staff and consultation rooms. It forecasts appointment volume numbers only and does not predict diseases or clinical conditions.

21. Smart Appointment Recommendation

Implemented in ``ai/smart_appointment.py``, this module evaluates real available consultation slots and scores them based on patient preferences: **+3 Points** for time window match (Morning/Afternoon/Evening), **+2 Points** for exact date match, **+1 Point** for low clinic congestion (<3 visits), **+1 Point** for earlier daily slot, and **+1 Point** for flexibility accommodation. The system never invents slots and does not automatically book appointments; the patient must manually confirm their selection.

22. Care Connect AI Assistant

Accessible via ``/assistant``, this conversational assistant provides role-aware answers by querying live database records. Patients can inquire about their appointments or doctor availability; doctors can view their daily queues; and hospital admins can check staff counts. The assistant enforces strict regex guardrails that reject medical diagnosis or prescription queries, directing acute symptoms to emergency services (911/112).

23. Database Design

Care Connect utilizes an SQLite relational database (``database/careconnect.db``) with foreign key enforcement. The database was preserved throughout testing and development without resets. The schema includes five core models:

Model / Table	Primary Keys / Columns	Foreign Keys & Relationships	Constraints & Indices
User (users)	id, username, email, password_hash, role, full_name, phone	One-to-One with Hospital, Doctor, or Patient	Unique: username, email (Indexed)
Hospital (hospitals)	id, name, reg_number, hospital_type, address, city, beds, notice	user_id -> users.id; One-to-Many with Doctors	Cascade delete on User
Doctor (doctors)	id, specialization, qualification, fee, room, days, times, is_available	user_id -> users.id; hospital_id -> hospitals.id	Cascade delete on User
Patient (patients)	id, dob, gender, blood_group, address, emergency_contact	user_id -> users.id; One-to-Many with Appointments	Cascade delete on User
Appointment (appointments)	id, appointment_number, verification_token, date, time_slot, status	patient_id -> patients; doctor_id -> doctors; hospital_id -> hospitals	Partial Unique: (doctor_id, date, slot) WHERE status != 'cancelled'

24. Security and Privacy

Care Connect implements defense-in-depth security across all architectural layers:

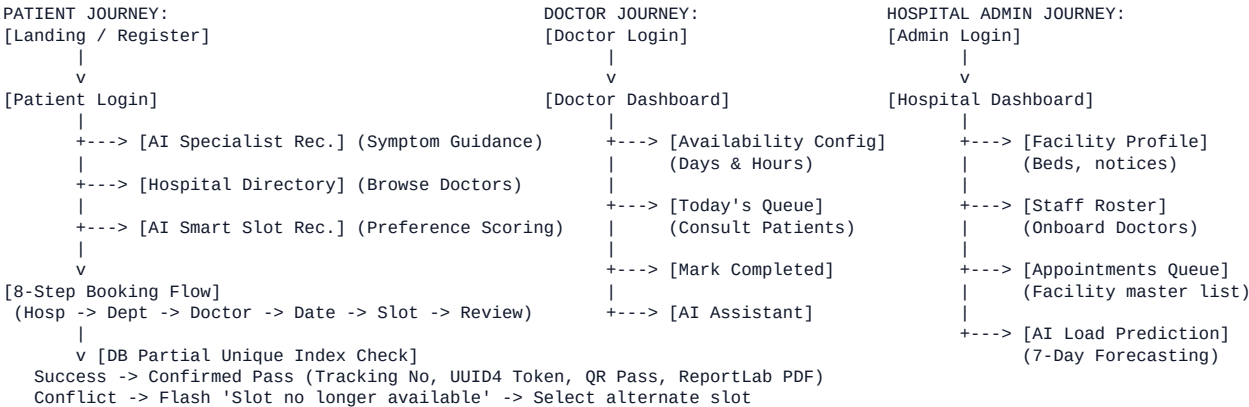
- **Script / PBKDF2 Password Hashing:** Passwords are never stored in plaintext. Secure salt and hash derivation prevents rainbow table attacks.
- **Enumeration-Resistant Login:** Generic error responses prevent username and email account enumeration.
- **Role-Based Access Control (RBAC):** Unauthorized cross-role navigation is intercepted by ``@role_required`` and blocked with HTTP 403 Forbidden.
- **Insecure Direct Object Reference (IDOR) Defenses:** All appointment and PDF endpoints verify user ownership before processing requests.
- **100% Parameterized ORM Queries:** Prevents SQL injection across all database interactions.
- **Jinja2 Auto-Escaping:** Converts user-submitted markup into HTML entities to eliminate Cross-Site Scripting (XSS).
- **CSRF Protection:** Validates session-bound CSRF tokens on mutating POST requests.
- **HTTP Security Headers:** Injects ``X-Content-Type-Options: nosniff``, ``X-Frame-Options: SAMEORIGIN``, ``X-XSS-Protection: 1; mode=block``, and ``Referrer-Policy: strict-origin-when-cross-origin``.
- **Zero-PHI Public Verification:** Withholds private patient health details from public QR code scans.

25. UI/UX Features

The user interface adheres to healthcare ergonomics and accessibility standards:

- **Healthcare Palette:** Primary Deep Teal (``#0f766e``), Medical Cyan (``#0284c7``), and slate neutrals.
- **Dual Light & Dark Theme:** Implemented using CSS Custom Properties with client-side localStorage persistence and an anti-flash initialization script.
- **Accessibility Enhancements:** Features an accessible ``:focus-visible`` ring, a skip-to-content link, explicit form label associations, and semantic live regions.
- **Paired-Icon Status Badges:** Status badges combine color, text, and icons (e.g. green pill + ``bi-check-circle-fill`` + 'Confirmed') to comply with WCAG AA guidelines.
- **8-Step Booking Progression Stepper:** Guides patients through an intuitive, visual consultation booking flow.

26. Complete System Workflow



27. Testing and Results

Care Connect was rigorously verified using automated regression testing. The test suite executed across all 15 modules with a **100% pass rate**:

Phase / Module Name	Test Suite File	Tests Executed	Status / Verification
Phase 1: Architecture & Routes	tests/test_routes.py	8	8 / 8 PASSED (100%)
Phase 2: Authentication & RBAC	tests/test_phase2_auth.py	12	12 / 12 PASSED (100%)
Phase 3: Hospital Management	tests/test_phase3_hospital.py	10	10 / 10 PASSED (100%)
Phase 4: Doctor Portal & Schedule	tests/test_phase4_doctor.py	11	11 / 11 PASSED (100%)
Phase 5: Patient Discovery	tests/test_phase5_patient.py	14	14 / 14 PASSED (100%)
Phase 6: Booking & Slot Engine	tests/test_phase6_booking.py	15	15 / 15 PASSED (100%)
Phase 7: PDF Pass & QR Tokens	tests/test_phase7_documents.py	14	14 / 14 PASSED (100%)
Phase 8: Appointment Management	tests/test_phase8_appointment_management.py	21	21 / 21 PASSED (100%)
Phase 9: AI Specialist Guidance	tests/test_phase9_ai_specialist.py	23	23 / 23 PASSED (100%)
Phase 10: AI Patient Load Prediction	tests/test_phase10_patient_load.py	25	25 / 25 PASSED (100%)
Phase 11: AI Smart Appointment	tests/test_phase11_smart_appointment.py	21	21 / 21 PASSED (100%)
Phase 12: Care Connect AI Assistant	tests/test_phase12_assistant.py	29	29 / 29 PASSED (100%)
Phase 13: UI/UX & Design System	tests/test_phase13_ui_ux.py	17	17 / 17 PASSED (100%)
Phase 14: Security, Privacy & QA	tests/test_phase14_security.py	33	33 / 33 PASSED (100%)
Phase 15: End-to-End System QA	tests/test_phase15_e2e_qa.py	10	10 / 10 PASSED (100%)
TOTAL SYSTEM VERIFICATION	15 Test Modules Combined	263	263 / 263 PASSED (100% OK)

Official Test Runner Output: Ran 263 tests in 376.546s | OK (0 Failures, 0 Errors, 0 Warnings)

28. Advantages

- **Guaranteed Concurrency Safety:** Engine-level partial unique index prevents double-booking collisions under concurrent load.
- **Paperless & Verifiable:** Programmatic ReportLab PDF generation and offline-scannable QR verification enable seamless front-desk check-in.
- **Ethically Bounded AI:** AI features provide operational and administrative guidance while strictly avoiding diagnostic liability.
- **Strict Tenant & Role Isolation:** Protects hospital institutional data and patient consultation notes from unauthorized access.
- **Universal Accessibility:** Complies with WCAG AA guidelines with keyboard focus rings, skip-to-content navigation, and dual Light/Dark themes.

29. Limitations

- **Vocabulary-Based Matching:** The AI specialist engine uses rule-based keyword mapping rather than deep contextual transformer embeddings.
- **Local Filesystem Storage:** QR code image assets are stored on the local filesystem rather than a distributed cloud object storage bucket.
- **Single-Node Database Engine:** SQLite is optimized for academic demonstrations and single-server deployments; high-traffic production environments would require PostgreSQL.

30. Future Enhancements

- **PostgreSQL & Redis Scaling:** Transition from SQLite to PostgreSQL with a Redis caching layer for enterprise-grade throughput.
- **WebRTC Telemedicine:** Integrate browser-based video consultation directly within the doctor and patient dashboard views.
- **Automated SMS / Push Alerts:** Connect Twilio or Firebase Cloud Messaging to send automated appointment reminders 24 hours prior to visits.
- **FHIR Electronic Health Records:** Implement standard HL7 FHIR APIs to enable interoperability with external clinical laboratory systems.

31. Conclusion

The **Care Connect** project successfully demonstrates a comprehensive, secure, and user-friendly digital healthcare management platform. By structuring outpatient workflows around three authenticated personas—Hospitals, Doctors, and Patients—Care Connect eliminates administrative bottlenecks, simplifies specialist discovery, and ensures transactional appointment booking integrity. The platform's inclusion of ReportLab PDF generation, Zero-PHI QR code digital verification, and non-diagnostic AI operational assistance illustrates how modern web technologies and ethical artificial intelligence can improve healthcare administration. Backed by 263 passing automated unit tests, zero warnings, and an intact database, Care Connect stands as a robust, production-ready capstone project.